



Astrophysicist

Task A: Being an astrophysicist

1) Fill in the gaps.

Astrophysics is a branch of astronomy.

Astronomers study the universe using *observational techniques* to assess and categorise various *celestial* bodies.

Astrophysicists have a more *narrowed* focus. They study the physics of the universe such as how the celestial *bodies* interact or other *physical* processes.

2) a) The table below shows the approximate average distances from Earth to some of the other planets in our solar system. Write the distances in standard form.

From	To	Distance (km)	Standard form
Earth	Mercury	91 691 000	9.1691×10^7
Earth	Venus	41 400 000	4.14×10^7
Earth	Mars	78 340 000	7.834×10^7
Earth	Jupiter	628 730 000	6.2873×10^8
Earth	Saturn	1 275 000 000	1.275×10^9

2) b) In 2021, the fastest speed recorded by a spacecraft was 163 km/s. Using that speed and the formula

$$\text{Distance} = \text{Speed} \times \text{Time}$$

To	Distance (km)	Time (days)
Mercury	91 691 000	7
Venus	41 400 000	3
Mars	78 340 000	6
Jupiter	628 730 000	45
Saturn	1 275 000 000	91

Calculate how long it would take, in days, to travel from Earth to each planet.

Round your answer appropriately to the nearest day.



Task B: Determining terrain features

1) Calculate the value (to 1 d.p.) of the unknown measurement in each case.

a) $\theta^\circ = 19^\circ$, $a = 56$ cm, $b = 19.3$ cm

b) $\theta^\circ = 56^\circ$, $a = 19$ cm, $b = 28.2$ cm

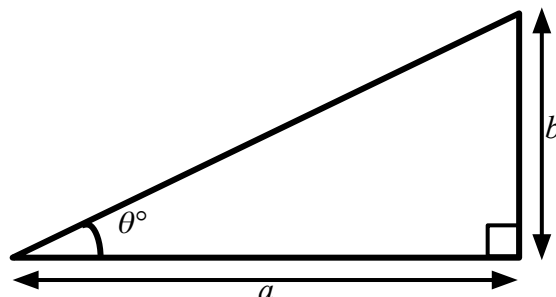
c) $\theta^\circ = 19^\circ$, $b = 56$ cm, $a = 162.6$ cm

d) $\theta^\circ = 56^\circ$, $b = 19$ cm, $a = 12.8$ cm

e) $a = 56$ cm, $b = 19$ cm, $\theta^\circ = 18.7^\circ$

f) $a = 19$ cm, $b = 56$ cm, $\theta^\circ = 71.3^\circ$

Not drawn
accurately



2) The wall of a crater on Mars is 1.2 km tall. When the wall casts a shadow with length 557 m, what angle (to 1 d.p.) is the sun at?

$$\tan^{-1}\left(\frac{1200}{557}\right) = 65.1^\circ$$

3) Another crater on Mars has a wall which casts a shadow of length 21 m. This occurs when the sun is at an angle of 67° . What is the height (to 2 d.p.) of this wall?

$$21 \times \tan(67) = 49.47 \text{ m}$$

4) The largest crater on the near side of the Moon is called Clavius. The walls of this crater are 6.6 km high.

Calculate three pairs of values for the angle of the sun and the corresponding length of the shadow cast by the crater wall.

There are infinitely many pairs that will work here. To check if your values are correct, perform this calculation:

$$\text{length of shadow} \times \tan(\text{angle of sun})$$

Your answer should round to 6.6 (if in km) or 6600 (if in m).

5) In order to calculate an estimate for the height of a crater wall, we have produced a model/sketch of the situation.

State an assumption that we have made that may affect the accuracy of our estimated value.

A possible assumption:

We assumed that the surface of the planet was flat when it is actually curved.

